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Time Series Analysis
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Time Series Methods for Forecasting Volatility

Abstract

Forecasting volatility is of paramount importance to quantitative financial analysis. Volatility is a critical component of options pricing and therefore of central importance in risk management.¹ Recently, techniques borrowed from machine learning have become more prominent² in time series forecasting of financial data. We seek to compare traditional time series forecasting models to more recent deep learning models. In doing so we employ an exponential smoothing model, a GARCH model, and an LSTM model. We conclude that for the stock analyzed, more naive methods tend to outperform deep learning methods.

I. Background

Volatility is a fundamental component in quantitative finance measuring the degree of variation in stock prices over time.³ It is a key parameter in risk assessment in the pricing of derivatives, particularly options. Traditionally, quantitative analysts have relied on models like the Generalized Autoregressive Conditional Heteroskedasticity (GARCH)¹ and Exponential smoothing to predict volatility.

Volatility is important to both investment managers as well as trading firms. Investment managers use volatility predictions to minimize portfolio risk, typically forecasting volatility over several months.¹ On the other hand, trading firms or hedge funds rely on volatility forecasts to price derivatives accurately. Often traders will forecast volatility over short periods such as days, hours, or even minutes. Traders who are able to predict volatility more accurately can exploit arbitrage opportunities gaining a competitive advantage.³

In recent years, the complex nature of volatility has sparked interest in the application of deep learning and neural networks. Particularly, models like a Long Short Term Memory (LSTM), which are able to capture temporal behavior in data, have shown promising results in

¹ Brooks, C. and Persaud, G. (2003), Volatility forecasting for risk management. *J. Forecast.*, 22: 1-22.
<https://doi.org/10.1002/for.841>

² Sullivan, Jason C. "Stock Price Volatility Prediction with Long Short-Term Memory Neural Networks." *Department of Computer Science, Stanford University, Stanford, CA*, jsull@stanford.edu.

³ Mancino, M. E., Recchioni, M. C., & Sanfelici, S. (2017). *Fourier-Malliavin Volatility Estimation: Theory and Practice*. Springer Cham. <https://doi.org/10.1007/978-3-319-50969-3>

high frequency predictions⁴. This study aims to extend research in the applications and accuracy of neural networks in volatility predictions when compared to more traditional statistical methods.

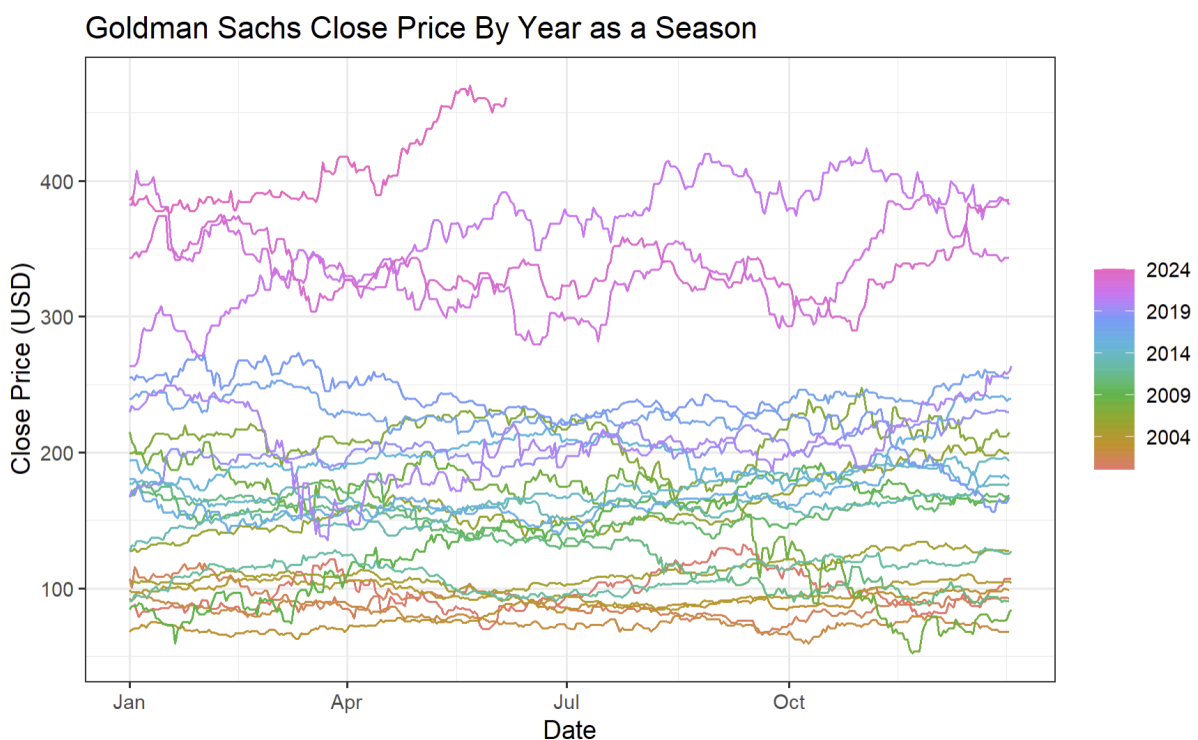
II. Data

Our study utilizes data from the Wharton Research Data Services (WRDS) platform.⁵ We choose to compare our models using Goldman Sachs daily close prices. This choice is fairly arbitrary, however, Goldman Sachs might minimize its idiosyncratic volatility via its diverse financial incentives, a virtue of its broader structure and market leadership in investment banking. If true, this makes Goldman Sachs an ideal candidate for testing volatility forecasting as the stock is uniquely representative of overall market trends, and still maintains individuality in a way that index funds or ETFs do not.

We query for daily high, daily low, close price, open price, and daily average price. From this data we construct log daily returns by dividing each close price by the previous close price, then taking the natural logarithm of this value. This constructs percent returns, then log scales them. Any return which represents a stock value decrease would then be a number less than 0, any return which represents an increase in stock price is greater than 0. Any period in which stock prices remain the same will have a return of 0.

Following the construction of log daily returns, we calculate a 21 day rolling standard deviation. This is synonymous with a rolling 21 day historical volatility. Given that standard deviation is synonymous with volatility, this rolling standard deviation represents historical volatility over the past 3 weeks (21 days). Rolling standard deviation will become our variable of interest for the rest of this study.

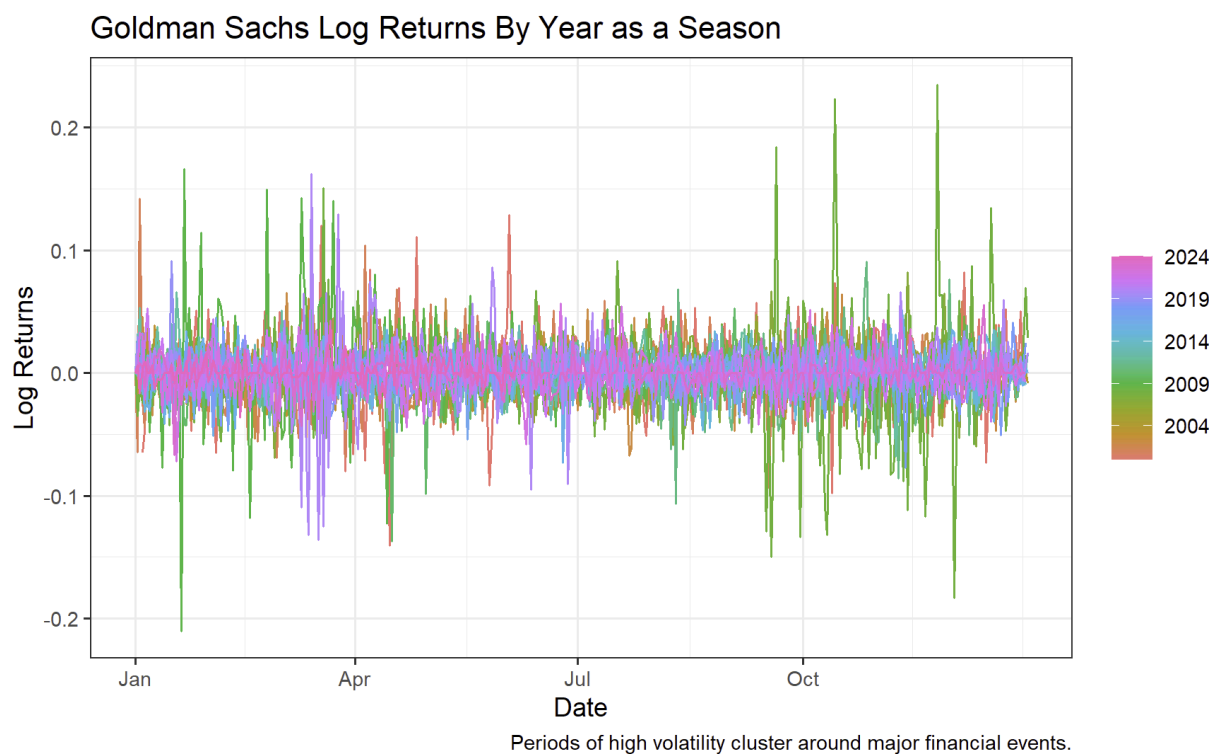
Figure 1.



We begin a cursory analysis of the given data by plotting close price and log return seasonality. Observing the seasonality plot of price by year as in figure 1, we note that there is no obvious seasonality in closing prices. Price appears to fluctuate stochastically, clearly increasing over time. Other than this expected result, there don't appear to be any notable features in the price structure of the data.

Next we investigate the seasonality of returns as seen in figure 2. We note here that this is logically equivalent to the first order difference. Returns are log scaled, meaning that the division operator becomes equivalent to the subtraction operator and our multiplicative differencing becomes additive. The first-differenced data is normally distributed and it appears to be stationary. Variance does increase in years like 2008, but these are unusual events and it is forgivable to have some heteroskedasticity apparent in such a financial crisis. Additionally, when implementing the GARCH model we will not seek to remove this heteroskedasticity, but rather model it. In this way, the slight heteroskedasticity exhibited in figure two is actually a desirable feature in our data.

Figure 2:



III. Method

A. Exponential Smoothing

We first seek to approximate volatility using exponential smoothing methods. Smoothing methods are common for volatility estimation in the world of high frequency trading.

Specifically, techniques from financial signal processing utilize Fourier transformations to represent financial data in the signal domain, then applying filters to denoise the data.³ That methodology is very powerful when estimating short term (instantaneous) volatility. Here, we adapt this intuition to achieve a similar result on a longer timescale. Rather than using Fourier-Malliavin smoothing, we apply exponential smoothing. This smoothing method is more appropriate for more discrete data.

We utilize exponential smoothing with additive errors, additive trend, and no seasonality to model volatility. Although there appears to be conditional heteroskedasticity in the volatility as aforementioned, this is not the same as seasonality, and as a result seasonality is not included. All models are forecast for a 6 month period, beginning in the start of 2024.

B. GARCH

The GARCH model is particularly useful for its ability to model conditional heteroskedasticity. Conditional heteroskedasticity describes a phenomenon in which some information within the data predicts the volatility of future data. In financial data, this is often manifest as volatility clustering. Whole periods of time tend to either be characterized by high volatility or low volatility. This can be seen in years such as 2008.

GARCH models require the definition of an ARMA model within their specification. For our model, we use an ARMA(0, 0) model. This is a mean-model; meaning that the ARMA predicts the forecast value to be the mean of the training data at all points. The GARCH component of the model requires two parameters, one for the number of squared past residuals to be included, and one for the influence of past conditional variances. We use a GARCH of order (2, 1) for our analysis. This is informed by the seminal literature surrounding volatility estimation with GARCH.⁶

C. LSTM

An LSTM is a type of neural network constructed to handle time series data. Neural networks are machine learning algorithms that capture patterns in data as it passes through its nodes. As the data is passed through the nodes, it is mapped through higher dimensions through the use of activation functions. It is passed through layers with weights and biases that are adjusted with respect to a “loss function” through a process called backpropagation.

LSTM’s unique architecture that handles temporal data consists of a cell state, hidden state, input gate, forget gate, and output gate. The cell state contains the long term memory, while the hidden state iteratively updates the cell state given the previous hidden state and the new inputs. The input gate decides information to add to the cell state while the forget gate manages

⁶ Xie, Haibin, and Chengtan Yu. "Realized GARCH Models: Simpler is Better." *Finance Research Letters*, vol. 33, 2020, 101221. ISSN 1544-6123. <https://doi.org/10.1016/j.frl.2019.06.019>. Accessed from <https://www.sciencedirect.com/science/article/pii/S1544612318308365>.

the flow of information out of the cell state. Finally, the output gate decides what to output into the new hidden state.

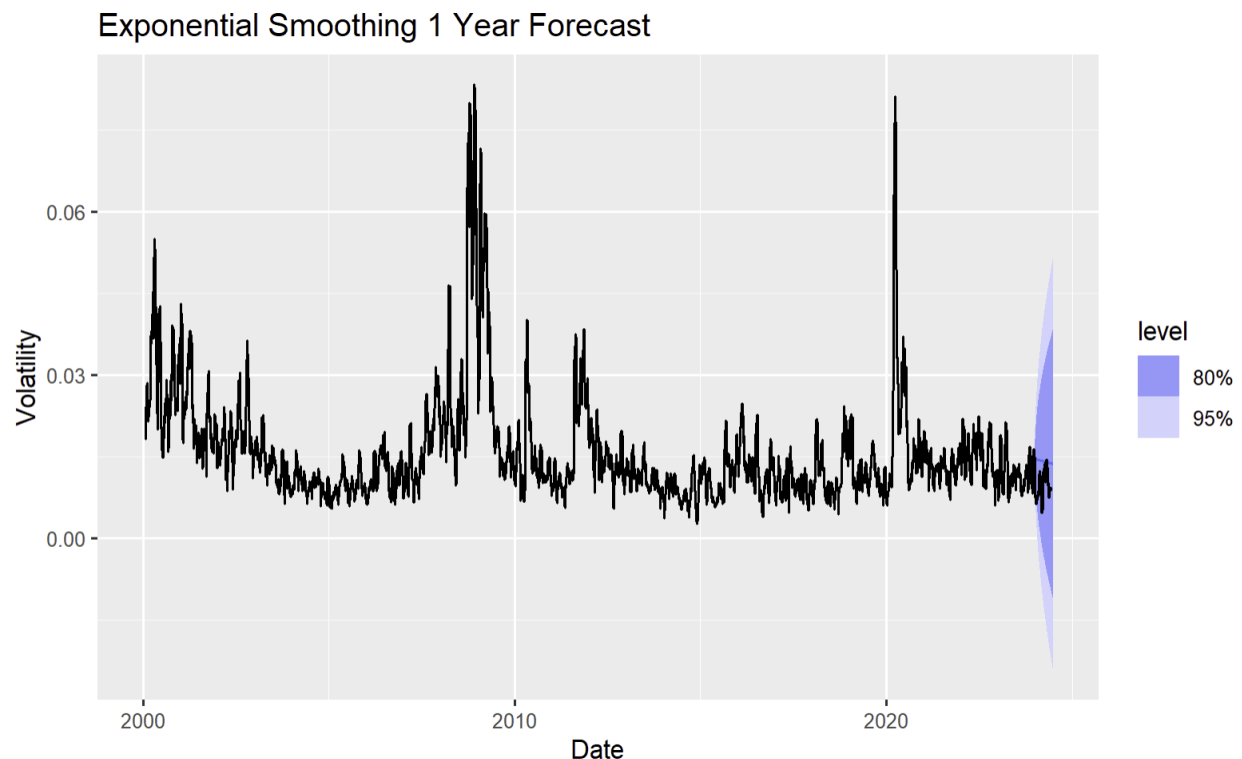
The architecture outlined in this paper follows a traditional LSTM setup with an LSTM layer with 128 nodes and a RELU activation function. Then the output layer consists of a dense layer with one node and a linear activation function. The model is trained on all features in the data set, with the target variable being the calculated volatility. The training set consists of 98% percent of the data capturing time periods from January 2000 to December 2023 while the test set consists of 2 percent of the data from January 2024 to June 2024. Finally we set epochs to 5 to capture a balance of underfitting and overfitting.

IV. Results

A. Exponential Smoothing

Despite delivering a fairly simple projection, the exponential smoothing model has strong explanatory power. With an RMSE of .005229... the model has a fairly low root mean square error. This perhaps reflects the fact that with highly stochastic data, a naive forecasting method often performs better than more complex methods.

Figure 3



B. GARCH

To evaluate the GARCH model, we first plot the model against the absolute value of returns over all of the data. The GARCH appears to successfully demonstrate the condition nature of the heteroskedasticity inherent in the data, as evidenced by how the standard deviation

tracks the magnitude of returns. This is an encouraging sign for our forecasts, and affirms the previous contention that conditional heteroskedasticity in returns might enhance the explanatory power of the GARCH model, or at least give the model something to explain beyond what exponential smoothing can.

Figure 4

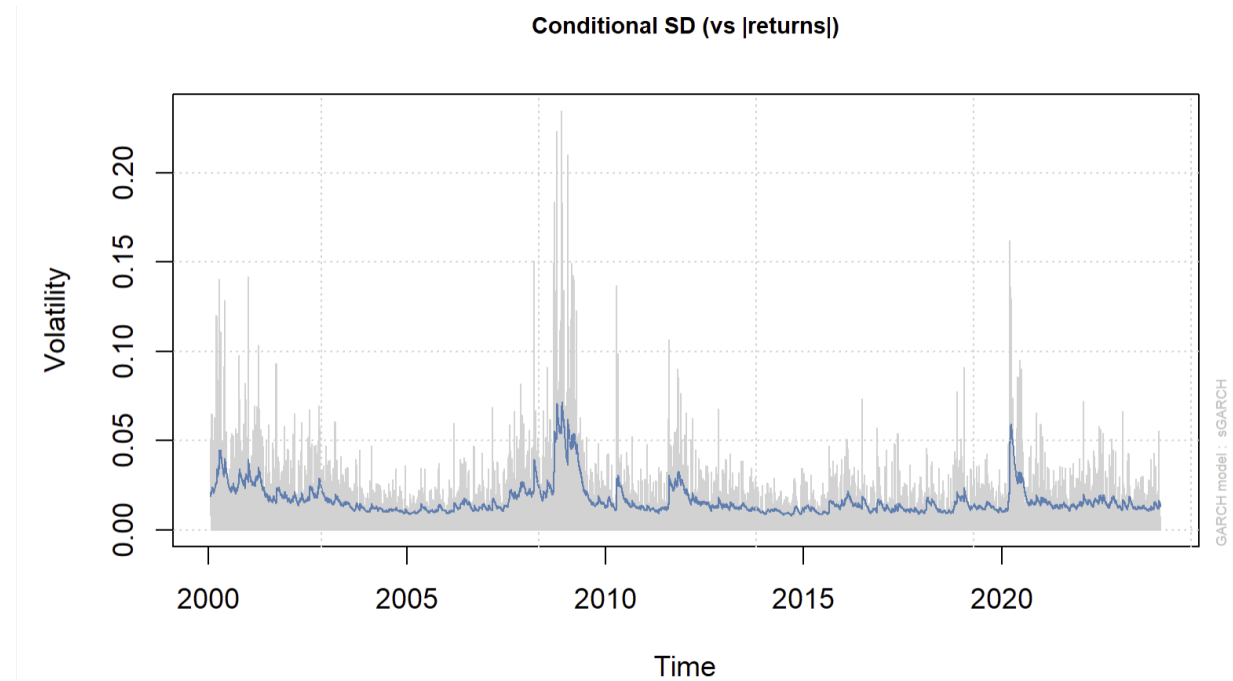
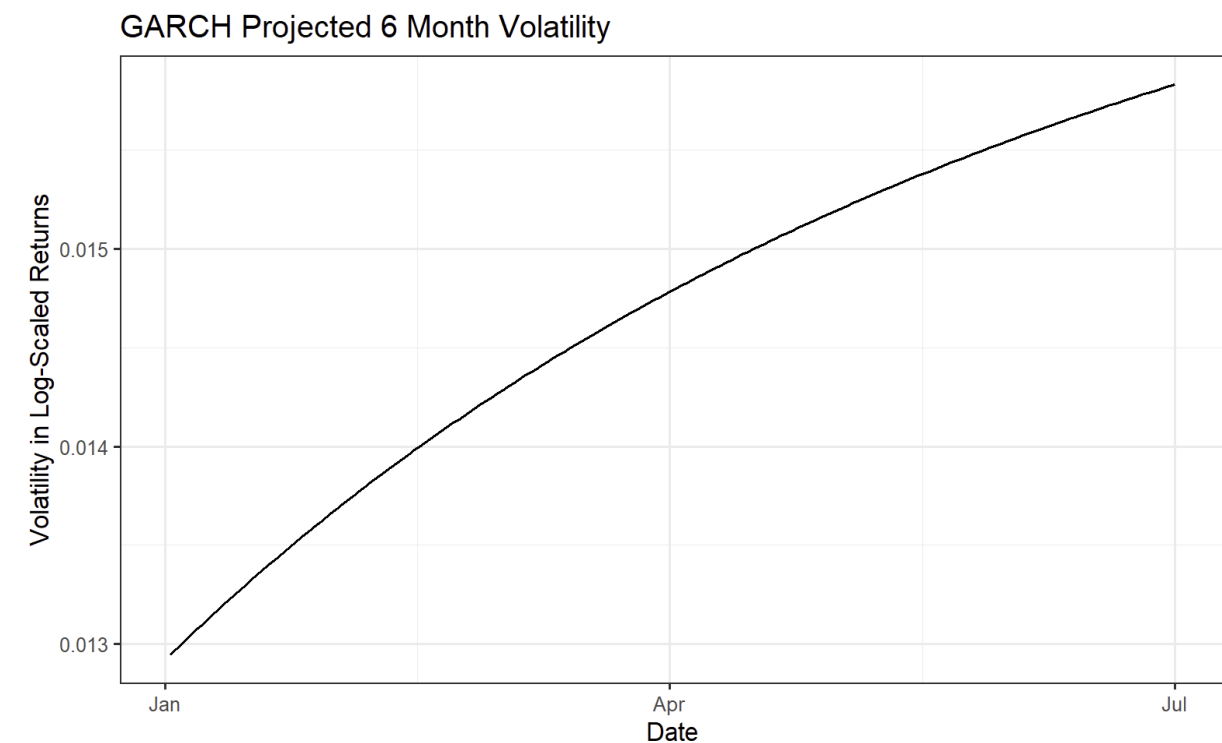


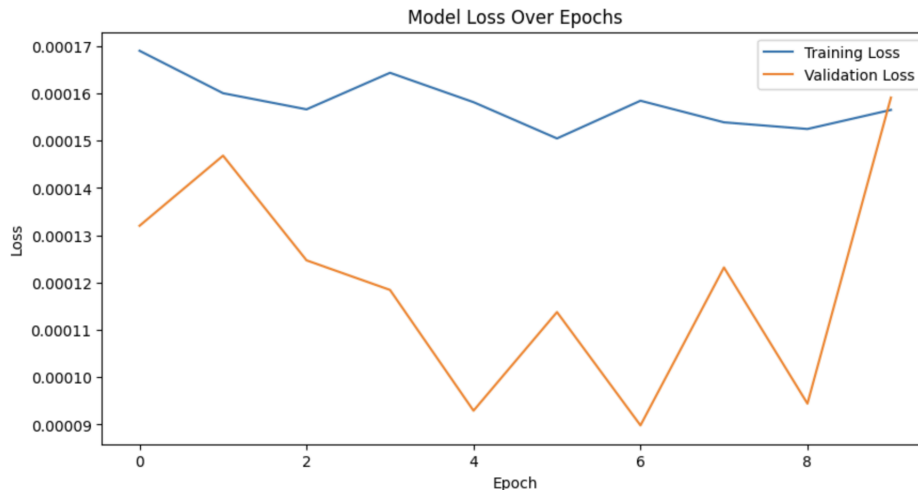
Figure 5.



Here we observe the predicted values of volatility for the next 6 months, as calculated by the GARCH model. Note, the increasing with marginal decrease form of this plot is a commonality among GARCH models. Again, despite being fairly simple, this model performs very well against the test data with an RMSE of .00542.

C. LSTM

Figure 6:



For the LSTM, we first plot the Loss (MSE) over a period of 10 epochs. As we see in the validation set, which represents unseen data, the Loss drops down and then starts to pick up around 10 epochs. This rise at the end represents overfitting where the model isn't able to generalize well on unseen data. Thus we set epochs to 5 to minimize the Loss we see in our test set.

Figure 7:

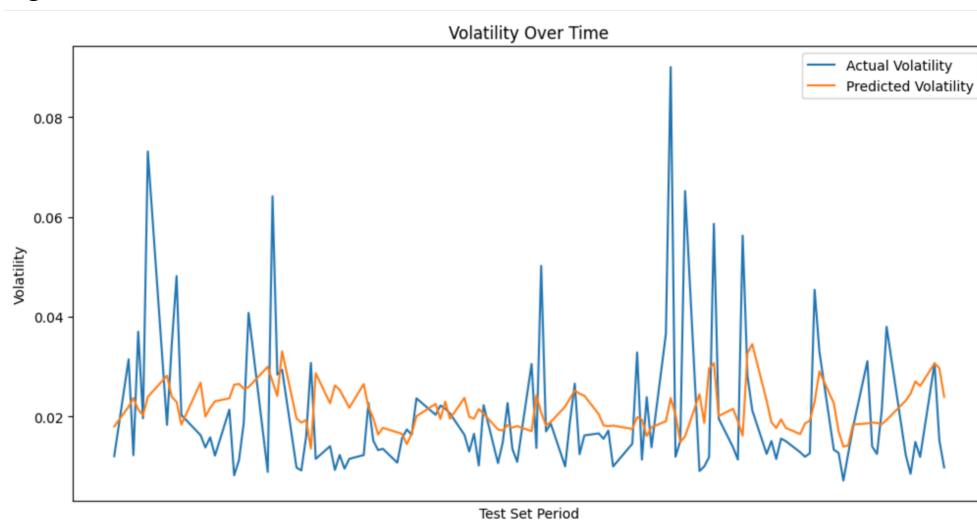


Figure 7 is a plot of the LSTM predicted volatility against actual volatility for our test set. Although the predictions appear to capture the underlying behavior of the data, the model struggles to pick up on the noisiness of the data. We reach an RMSE of 0.015 indicating poor performance when compared to traditional models.

V. Limitations

This study was subject to a few limitations. First, the scope of the study is limited in nature as it only examines one stock. It is not obvious that the trends found here are applicable to all stocks. This requires further investigation. Additionally, using GARCH models in the R programming language is very difficult as the available packages for GARCH modeling do not work well with other commonly used time series packages. A deeper and more tempered exploration of conditional heteroskedasticity could be achieved with more facile tools. Additionally, further research might benefit from unique data coercion. GARCH models tend to perform better with higher frequency data, and therefore the rolling standard deviation approach to volatility may not have been the best for optimizing specifically with respect to the GARCH. When considering the LSTM, we find that further fine-tuning can be made to improve its predictions. For example, employing regularization techniques, adjusting learning rates, and adding or removing layers, may significantly change the predictions seen. Furthermore, literature suggests the combining of two machine learning algorithms may further increase performance in time series predictions.⁷

VI. Conclusion

For our test set the traditional time series approaches seemed to outperform the LSTM. This reflects the stochastic nature of stock price and volatility data, and reflects that oftentimes the best predictor of stock price is the naive prediction. Additionally, it suggests that the complexity introduced by LSTM models may not necessarily translate to better performance in financial time series forecasting, where simpler models often capture the underlying patterns more effectively. Moving forward research ought to investigate the efficacy of models which utilize a combination of tradition and deep learning models.

⁷ Abbasimehr, H., Paki, R. Improving time series forecasting using LSTM and attention models. *J Ambient Intell Human Comput* 13, 673–691 (2022). <https://doi.org/10.1007/s12652-020-02761-x>